Designed by Remus Teodorescu 2026

LECTURE 10.2

Electrolysers

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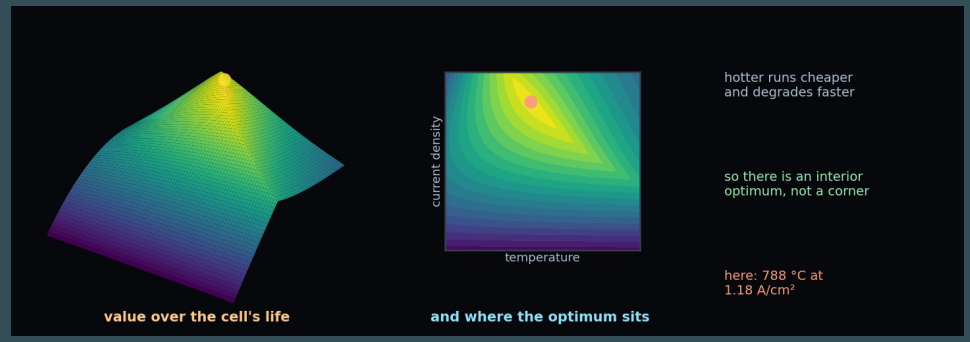
Deep Learning for Engineering · 2026

STEAM · VOLTAGE · LIFETIME



Schematic: the value a cell produces over its lifetime against temperature and current, with its optimal point in the interior - hotter is cheaper per hour and ages faster. The lecture sets out the electrolyser and the thermoneutral voltage, solves the fuel channel, and finds the optimal voltage by a gradient through one PINN trained over the whole operating range.

Learning Objectives

SECTIONS, IN ORDER

1. **The cell** — three electrolysers, the losses, the thermoneutral voltage

2. **The channel** — steam, current and heat settling together

3. **The optimal voltage** — one PINN for every operating point, the cost and its gradient

The physics is carried by the exercise itself; Liu carries the optimisation through a network. The example is Ex_10.2's electrolysis cell: the channel of L9.2, now with its electrode. This is the last physics lecture, and the place Part 2 has been heading.

READ, AND WHAT EACH ANSWERS

Ex_10.2 sections 3 – 5	where do the losses sit?
Ex_10.2 sections 1 – 3	how does the steam run short?
Liu ch. 5	what does a gradient buy?

Three Electrolysers

Water electrolyser — a cell that splits water into hydrogen and oxygen with electricity. The three industrial types differ in their electrolyte, and so in the ion that carries the charge and in the temperature at which they run.

ONE ELECTROLYTE EACH

alkaline — a potassium hydroxide solution near 80 degrees; OH^- carries the charge and a diaphragm keeps the gases apart

PEM — a humid polymer membrane near 70 degrees; H^+ crosses to the hydrogen side and liquid water is split at the oxygen side

solid oxide — a ceramic near 800 degrees; O^{2-} crosses to the air side, steam is split and heat supplies part of the energy

COST AND CONTROLLABILITY

	alkaline	PEM	solid oxide
temperature [°C]	70 – 90	50 – 80	700 – 850
energy [kWh/kg H_2]	50 – 78	50 – 83	45 – 55
cost [USD/kW]	500 – 1000	700 – 1400	stack >2000
minimum load [%]	10 – 30	10 – 40	about 50
ramp [% per s]	0.5 – 5	3 – 10	about 0.2
start from cold	minutes	minutes	hours: kept hot

Typical values, rounded; costs are IRENA's 2020 system prices (installed plants more: 2000 – 2600 USD/kW outside China, 600 – 1200 in China; IEA 2025). The solid oxide rows are one product's data sheet, and its limits are thermal, not electrical: the stack follows the current within seconds, and designs that manage the heat, such as Dynelectro's AC:DC cycling, reach low minimum load.

IRENA (2020); Dorn et al. (2026); IEA (2025)

Electrolysis at Industrial Scale

Power-to-X — the conversion of renewable electricity into hydrogen by electrolysis, and of the hydrogen into a fuel or chemical, such as methanol, that can be stored and shipped.

KASSØ, DENMARK, SINCE 2025

electrolysers	52 MW of PEM, from Siemens Energy
power	a 304 MW solar park beside the plant
product	up to 42 000 t of e-methanol a year

INSTALLED WORLDWIDE, AND 2030

installed	0.6 GW in 2021, 2 GW in 2024, 4.2 GW in 2025
by technology	in 2022: alkaline 60 %, PEM 30 %, SOEC under 1 %
by 2030	26 GW committed; 520 GW announced, much now out of reach
China	65 % of the capacity installed or committed

Laura Mærsk, e-methanol fuel
Jens Dohrn, CC BY-SA 4.0

2 MW PEM module, ITM Power
Bubble60, CC BY-SA 4.0

WHY ALKALINE STILL LEADS

mature	a century in industry
cheap	no noble metals, large stacks
China's makers	alkaline at 600 – 1200 USD/kW

Elsewhere 2000 – 2600 USD/kW (IEA). More: Kassø, ITM Power.

The Solid Oxide Electrolyser

Solid oxide electrolysis cell — a ceramic electrochemical cell that splits steam into hydrogen and oxygen near 800 degrees. The name comes from its electrolyte, a solid oxide (yttria-stabilised zirconia) that conducts only O^{2-} ions. In reverse it is a solid oxide fuel cell; the electrodes are therefore the fuel and air electrode, as anode and cathode swap between modes.

WHY THE HIGH TEMPERATURE PAYS

$$\Delta H = \Delta G + T\Delta S, \qquad \Delta G \downarrow \text{ as } T \uparrow$$

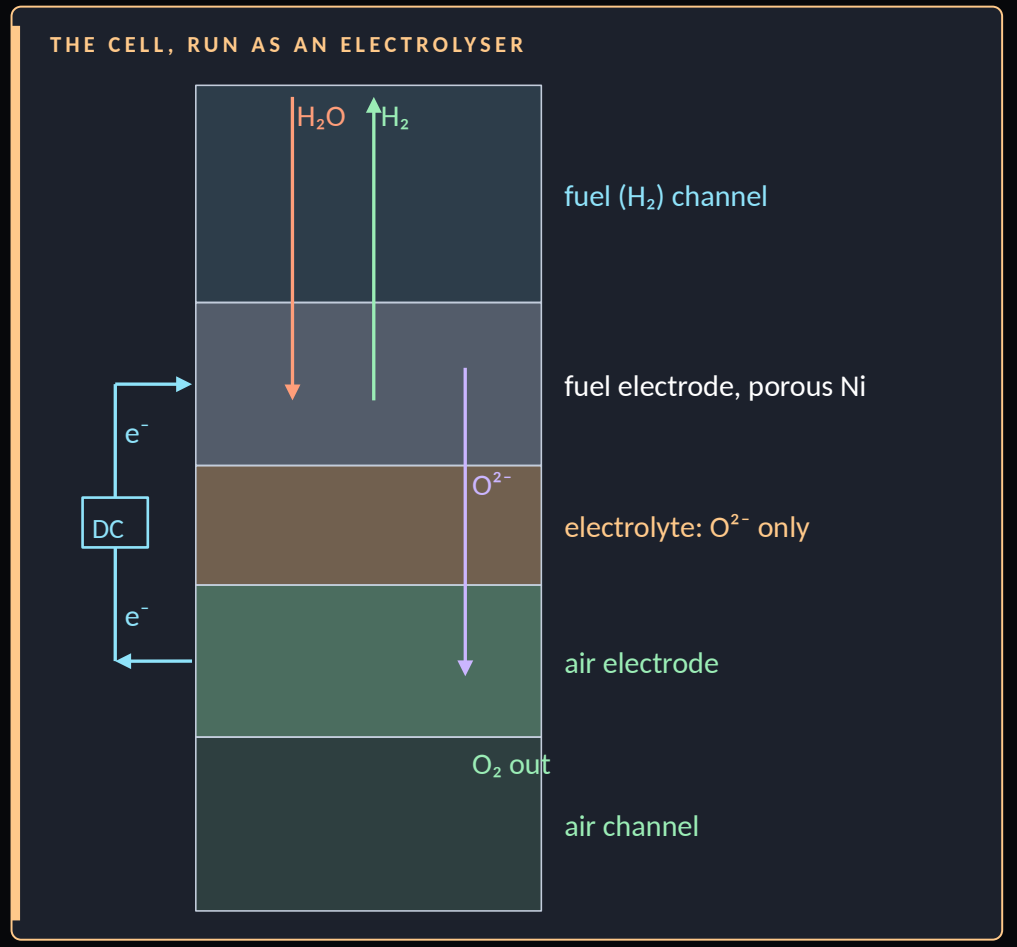
ΔH the energy to split water, ΔG the part that must be electricity, $T\Delta S$ the part heat can supply

THE THREE LAYERS, AND THEIR REACTIONS

fuel electrode — $H_2O + 2e^- \rightarrow H_2 + O^{2-}$: two electrons from the supply; the oxide ion enters the ceramic

electrolyte — O^{2-} hops through oxygen vacancies in the hot ceramic; electrons and gases cannot pass

air electrode — $2\ O^{2-} \rightarrow O_2 + 4e^-$: the ions return their electrons to the circuit and leave as oxygen



High Against Low Temperature

Why the heat pays — splitting a mole of water takes $\Delta H = \Delta G + T\Delta S$: the part ΔG must be electricity, the part $T\Delta S$ may enter as heat. Raising the temperature from 80 to 700 degrees lowers ΔG by about 15 %, so the reversible voltage $\Delta G/2F$ falls by about 0.18 V.

THE THREE NUMBERS, READ TOGETHER

ΔG falls 15 % — the real gain: E_{rev} falls from 1.18 to 1.00 V

$T\Delta S$ barely moves — T is 2.8× higher but ΔS much smaller: steam → gases, not liquid → gases

ΔH looks lower — the ≈ 41 kJ/mol that raised the steam left the reaction - still paid, outside it

and that heat — can be waste or process heat: the SOEC's advantage

80 °C AGAINST 700 °C, PER MOLE

quantity	80 °C PEM, liquid	700 °C SOEC, steam
ΔH [kJ/mol]	≈ 284	≈ 248
ΔG [kJ/mol]	≈ 228	≈ 194
$T\Delta S$ [kJ/mol]	≈ 56	≈ 54
$E_{rev} = \Delta G/2F$ [V]	≈ 1.18	≈ 1.00

Standard thermodynamic values, rounded: liquid water fed at 80 °C, steam at 700 °C. The lower reversible voltage is the efficiency gain; the evaporation heat is paid outside the stack.

The Polarisation Curve

Nernst potential — the voltage at which no current flows: the reversible voltage of the reaction, set by the local gas composition and the temperature; it rises as the steam is used up. The polarisation curve is this potential plus the losses, which grow with the current.

THE NERNST POTENTIAL, AND THE LOSSES ON TOP

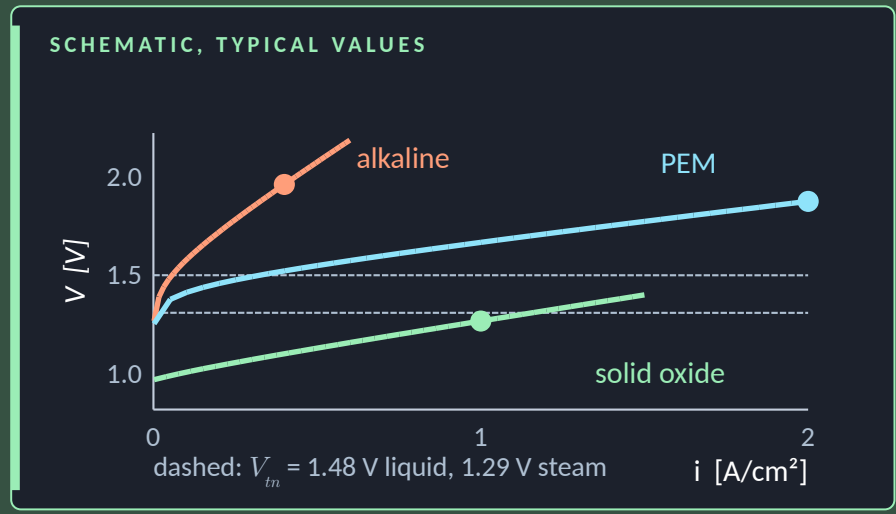
$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{(1-y)\,p_{O_2}^{1/2}}{y}\right)$$

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

E the Nernst potential, $E^0(T)$ the reversible voltage with every gas at 1 bar, y the steam fraction p_{O_2} the air side's oxygen, V the cell voltage, R the gas constant, T the temperature, $2F$ two electrons' charge η_{act}, η_{ohm}, η_{conc} the activation, ohmic and gas-transport losses; in electrolysis they add

As the battery of L10.1: there the open-circuit curves at the surface state of charge, set by diffusion, plus the reaction overpotentials - here the Nernst potential at the local gas, plus the losses above.

THREE CELLS: VOLTAGE TO ENERGY PER KG



alkaline	1.94 V at 0.4 A/cm²: 52 kWh/kg
PEM	1.85 V at 2.0 A/cm²: 49 kWh/kg
solid oxide	1.25 V at 1.0 A/cm²: 33 kWh/kg
per volt	26.6 kWh/kg (stack); system 45 – 55

The Thermoneutral Voltage

Thermoneutral voltage — the operating voltage at which the heat the losses release equals the heat the reaction absorbs, so the net heat is zero. For steam near 800 degrees it is 1.29 volts; stacks are operated at or near it, and with outside heat a stack can run below it, endothermically.

THE VOLTAGE FROM THE ENTHALPY, THE HEAT IN ONE TERM

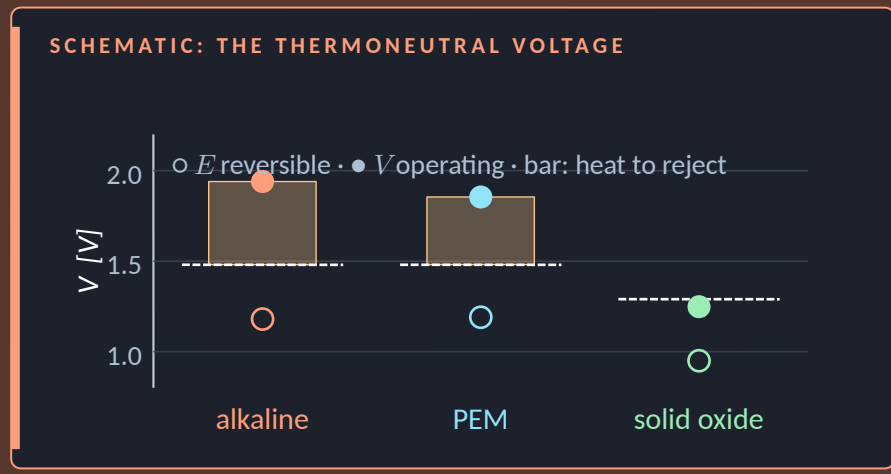
$$V_{tn} = \frac{\Delta H}{2F} = 1.29\ \text{V} \quad (\text{steam},\ 800\ ^\circ\text{C})$$

$$q = i(V - V_{tn})$$

V_{tn} the thermoneutral voltage, ΔH the enthalpy of splitting a mole of steam
$2F$ the charge of its two electrons (F Faraday's constant), q the heat per electrode area, i the current density

TALK · 2 MIN How would you explain, in your own words, where the heat of an electrolyser comes from and where it goes? Then, in the drawing on the right, why do the two cold cells reject heat while the solid oxide cell runs at its dashed line?

THREE CELLS ON ONE VOLTAGE AXIS



alkaline, PEM 0.4 – 0.5 V above: heat to remove

solid oxide at V_{tn}: no net heat in the stack

Liquid water: 1.48 V includes evaporation. Steam at 800 °C: 1.29 V, the steam raised outside.

The Fuel Channel

Fuel channel — the hydrogen channel of L9.2, run as an electrolyser: 90 % steam enters at 1 m/s and the electrode takes steam and returns hydrogen, so the steam mole fraction y, the share of steam molecules in the gas, falls along the channel.

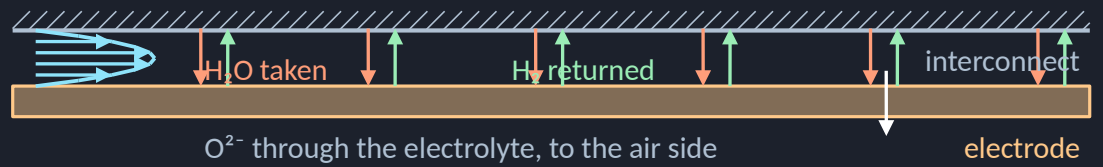
THE STEAM BALANCE AND THE ENERGY BALANCE

steam $\dfrac{dy}{dx} = \dfrac{1}{Pe}\dfrac{d^2y}{dx^2} - K_I\, i, \qquad Pe = 125$ — carried by the flow, taken by the electrode

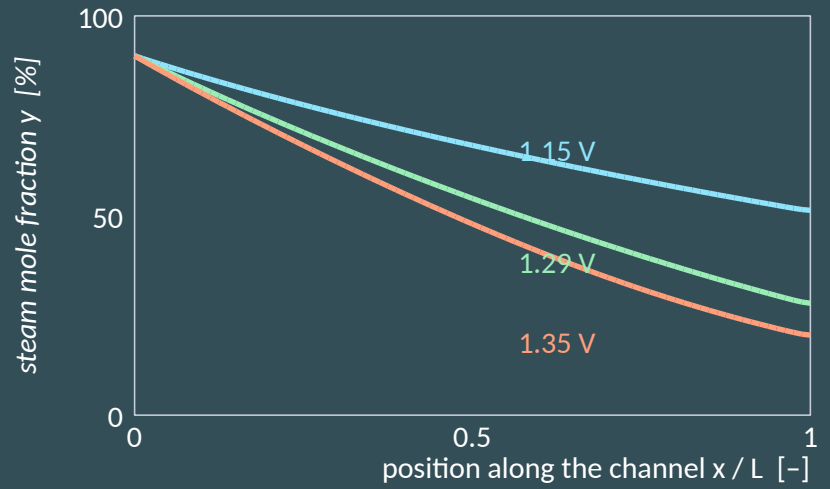
energy $\dfrac{d\theta}{dx} = \dfrac{1}{Pe_T}\dfrac{d^2\theta}{dx^2} + S_Q\, i(V - V_{tn}), \qquad Pe_T = 16$ — the current's heat warms or cools the gas

y the steam mole fraction, $\theta = (T - T_{in})/50$ K the scaled temperature, i the local current density [A/cm^2]
Pe, Pe_T the Péclet numbers; K_I, S_Q gather u, h, c, F, Γ, L; $V_{tn} = 1.29$ V
$i > 0$ in electrolysis, so $-K_I\, i$ makes y fall; hydrogen is $1 - y$, exchanged one for one : no equation of its own

STEAM TAKEN, HYDROGEN RETURNED



STEAM ALONG THE CHANNEL



THE CHANNEL IN THE THREE CELLS

	fed	the channel must
alkaline	KOH solution, in excess	carry gas bubbles out
PEM	liquid water, in excess	carry bubbles, cool
solid oxide	steam, 69 % used	keep steam to the outlet

The Local Current

Local current density — the current per electrode area at one point. The cell is held at one voltage V; the local steam fraction and temperature set the Nernst potential, the speed of the electrode reaction and the resistance, and the current is whatever satisfies the charge transfer there.

FROM THE POLARISATION CURVE, TERM BY TERM

the curve	$V = E(y,T) + \eta_{act} + \eta_{ohm} + \eta_{conc}$	at the local gas and temperature
activation	$\eta_{act} = V_T \operatorname{asinh}(i/2i_0)$	the voltage that drives the reaction at rate i
ohmic	$\eta_{ohm} = i\,ASR(T)$	Ohm's law; the resistance falls with T
gas transport	$\eta_{conc} = -(V_T/2)\ln(1 - i/i_L(y))$	the leaner gas at the electrode surface

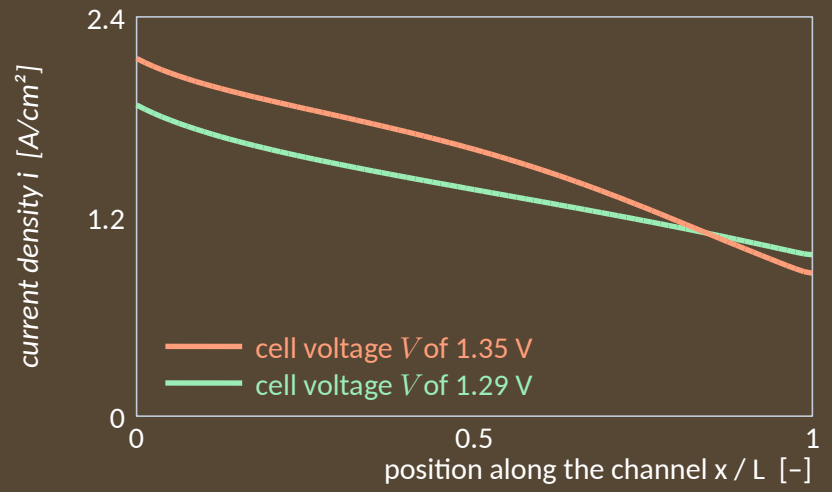
solve for i

$$V = E(y,T) + V_T \operatorname{asinh}\frac{i}{2i_0} + i\,ASR(T) - \frac{V_T}{2}\ln\left(1 - \frac{i}{i_L(y)}\right)$$

$E(y,T)$ the Nernst potential of the local gas, $V_T = RT/F$ the thermal voltage of L8.1, 93 mV at 800 °C
i_0 the exchange current density, $ASR(T)$ Arrhenius (estimates), $i_L(y)$ the limiting current, proportional to y; $i < i_L$
one unknown i in one monotone equation : Newton's method finds the root at every point of the channel

coupled — the current is set by the local y and T and in turn drains the steam and releases the heat, so the three equations are solved together.

THE CURRENT ALONG THE CHANNEL



THE CURRENT IN THE THREE CELLS

	along the channel	set by
alkaline	nearly even	gas bubbles
PEM	nearly even	bubbles, membrane
solid oxide	1.87 to 0.97 A/cm²	steam used up

Finite Volumes Along the Channel

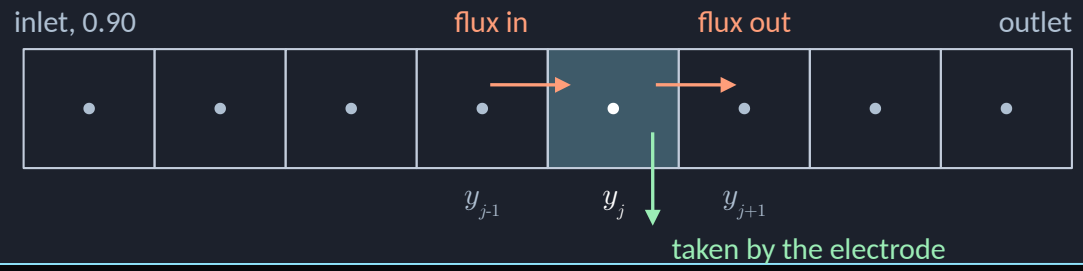
Finite volumes (FVM) — the channel cut into cells, and for each cell the balance of its steam and of its heat: what flow and diffusion carry in through one face, less what leaves by the other, equals what the electrode takes or the current releases. Shared faces conserve both exactly.

THE BALANCE OF CELL j

$$\frac{y_j - y_{j-1}}{h} - \frac{y_{j+1} - 2y_j + y_{j-1}}{Pe\,h^2} + K_I\, i_j = 0, \qquad \text{and its twin for } \theta_j, \qquad j = 1, \ldots, n$$

y_j, θ_j steam and scaled temperature in cell j, $h = 1/n$, i_j the root of the charge transfer in cell j
$2n$ equations, solved together by Newton-Raphson; the coupling blocks carry $\partial i/\partial y$ and $\partial i/\partial \theta$

EIGHT OF THE n CELLS, CELL j PICKED OUT



WORST ERROR AGAINST THE REFERENCE

cells	error in y [–]	error in T [K]	
25	0.0131	0.55	
50	0.0052	0.29	meets 0.01, 0.5 K
100	0.0024	0.14	
200	0.0015	0.07	

Each row is one grid: the largest error along the channel in the steam mole fraction and in the temperature, against the reference at 1.35 V, 800 °C. Fifty cells meet the agreed 0.01 and 0.5 K; the error falls about as the cell width (the flow term takes the upstream cell). One coupled solve: some 25 ms, repeated for every operating point.

One PINN for Every Operating Point

Parametric PINN — a PINN trained over three domains at once: the position x along the channel, the cell voltage V from 1.15 to 1.35 V and the inlet temperature from 750 to 850 degrees. After this one training it returns all three fields at any operating point in that range.

THE PHYSICS IN THE TRAINING

steam balance

$$\frac{\partial y}{\partial x} = \frac{1}{Pe}\frac{\partial^2 y}{\partial x^2} - K_I\, i$$

energy balance

$$\frac{\partial \theta}{\partial x} = \frac{1}{Pe_T}\frac{\partial^2 \theta}{\partial x^2} + S_Q\, i(V - V_{tn})$$

charge transfer

$$V_{cell}(i, y, T) = V, \qquad i = i_L(y)\,\sigma(\mathcal{N}_3)$$

boundaries

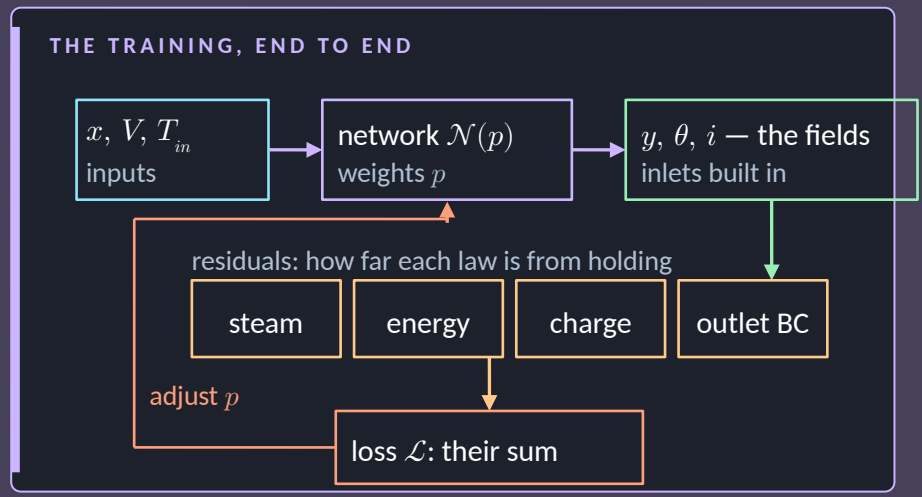
$$y = 0.90 + x\,\mathcal{N}_1, \quad \theta = x\,\mathcal{N}_2; \qquad \frac{\partial y}{\partial x} = \frac{\partial \theta}{\partial x} = 0 \text{ at } x = 1$$

$$\mathcal{L}(\theta) = \overline{r_y^2} + \overline{r_\theta^2} + \overline{\left(\frac{V_{cell} - V}{50\ \text{mV}}\right)^2} + \lambda\,\overline{\text{outlet slopes}^2}$$

r_y, r_θ the residuals of the two balances at points (x, V, T_{in}) over 1.15 to 1.35 V and 750 to 850 °C
the charge residual in units of 50 mV, so millivolts weigh like the balances; λ the outlet-slope weight

WHAT THE ONE TRAINING BUYS

one point	FVM in 25 ms is the tool
1000 points	one batch, about 500 times faster
then	the cost $C(V)$ and dC/dV
accuracy	0.0022, 0.19 K; FVM 0.0068, 0.40 K
offline	trained once, 33 min on a CPU



Optimisation of the Cell Voltage

Cost function — the number the operator minimises: electricity bought, less hydrogen sold, plus the share of the stack aged in the hour, as a function of the cell voltage V. The prices and the ageing law are assumptions; the optimum is conditional on them.

THE COST, ITS UNITS, AND ITS MINIMUM

$$C(V) = p_{el}\frac{V\,I}{1000} - p_{H_2}\frac{3600\,I\,M_{H_2}}{2F} + \frac{c_{stack}\,P_{rated}}{\Delta ASR_{EOL}}\Delta ASR_h$$

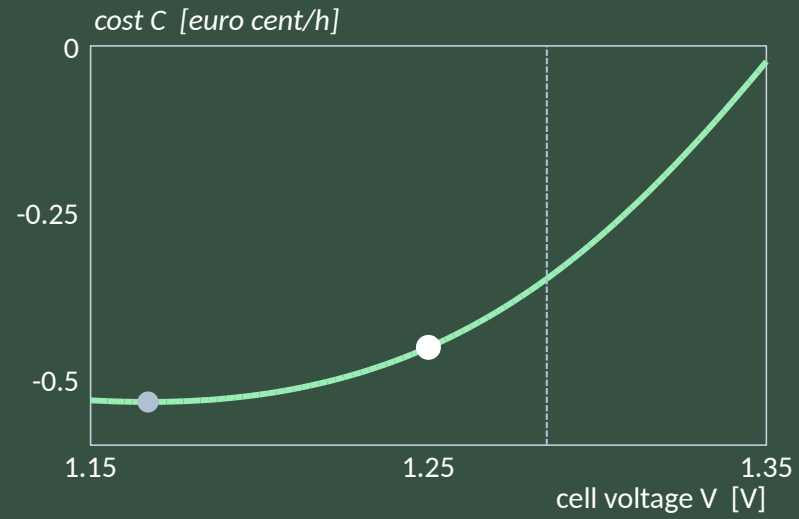
$$I = A\int_0^1 i\,dx \;\; [\mathrm{A}], \qquad \frac{V\,I}{1000}\;[\mathrm{kW}], \qquad \frac{3600\,I\,M_{H_2}}{2F}\;[\mathrm{kg/h}], \qquad V^* = \arg\min_V C \;\; \text{s. t.} \;\; \max_x |T - T_{in}| \le 10\ \mathrm{K}$$

C [cent/h] (negative : a profit); $p_{el} = 0.05$ euro/kWh, $p_{H_2} = 5$ euro/kg : assumptions; the 1000 converts W to kW the 3600 converts per second to per hour; ΔASR_h the resistance grown in the hour (the assumed law below)

THE OPTIMAL VOLTAGE IN PYTORCH

```
V = torch.tensor(1.25, requires_grad=True)   # the decision variable
J = C(V) + mu * relu(dT(V) - 10.0)**2        # the cost, the limit as a penalty
J.backward(); V -= lr * V.grad               # downhill through the network
```

THE COST AGAINST THE VOLTAGE



free optimum	1.167 V, but the cell cools 25 K
with the limit	V^* = 1.250 V, 0.45 cent/h earned
checked by	a sweep of 81 solver runs

No Optimal Point Over a Lifetime

Degradation — the slow rise of the cell's area-specific resistance over thousands of hours, from changes in its layers (right). The rate law below is an empirical form; its constants in Ex_10.2 are assumed, and the grown resistance only prices the hour - it is not fed back into the model.

THE DEGRADATION LAW, AND THE LIFETIME OBJECTIVE

$$\frac{dASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right) |i|^n$$

$$\min_{V(t),\,T(t)} \int_0^{t_{life}} [p_{el} P - p_{H_2} \dot{m}_{H_2}]\, dt \quad \text{s.t.} \quad ASR(t_{life}) \le ASR_{\max}, \quad |\nabla T| \le |\nabla T|_{\max}$$

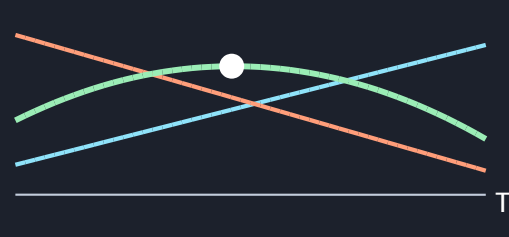
ASR the area-specific resistance [Ω cm^2]: the ohmic voltage lost per unit of current density
k_0, E_d, n ASSUMED in Ex_10.2, of the right order (published rates span an order of magnitude)
an empirical form, Arrhenius in T and a power of i, not derived from the mechanisms on the right

TALK · 2 MIN How would you explain, in your own words, why a device like this has no best operating point? Then, in the drawing on the right, what moves the white dot when the price of electricity falls?

WHAT AGES, AND WHY

layer	mechanism
fuel electrode	nickel grains coarsen and migrate: fewer reaction sites
air electrode	oxygen pressure at the interface lifts it off
steel, air side	chromium vapour deposits and blocks sites

THE TRADE, SCHEMATIC



cent/h: hotter is cheaper

hours left: fewer

product: an optimum inside

so the answer a trajectory $V(t), T(t)$

Operating for Cost and Lifetime

Operating strategy — the cell voltage $V(t)$ and temperature $T(t)$ chosen over the life of the stack to minimise the cost of each kilogram of hydrogen: the electricity bought hour by hour, and the stack replaced when its resistance reaches the end-of-life limit.

THE LEVELISED COST OF HYDROGEN

$$\mathrm{LCOH} = \frac{C_{stack} + \sum_t p_{el}(t)\, P(t)\, \Delta t}{\sum_t \dot{m}_{H_2}(t)\, \Delta t} \quad \text{over the stack's life}$$

LCOH the cost of a kilogram of hydrogen, C_{stack} the stack, replaced at end of life, $p_{el}(t)$ the hourly price $P(t)$, $\dot{m}_{H_2}(t)$ power and hydrogen at the chosen $V(t), T(t)$; the life ends when ASR reaches $ASR_{\max}$

THE TRADE-OFFS

higher voltage	more hydrogen per hour, faster ageing
higher temperature	lower resistance, faster ageing
cheap hours	run harder when the price is low
near V_{tn}	small heat, small gradients in the ceramic

WHAT THE PINN ADDS

one hour	the cost in a millisecond
gradient	dC/dV and dC/dT by autograd
a solver	one solve per variable per step
a lifetime	an optimiser over $V(t), T(t)$
Ex_10.2	seconds, against a sweep of 81 solves

```
C = lcoh(net, V_t, T_t)           # cost over a life
V_t = V_t - lr * grad(C, V_t)[0]  # schedule, downhill
```

Degradation rates are uncertain tenfold: a predicted lifetime ranks strategies, it is not a promise.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
across the channel	the gas profile, the porous electrode	Incropera ch. 14	the 1-D model: +5 – 9 % current; MP10.2A
thermal transients	inertia between operating points	Incropera ch. 5	cycling, start-up, quasi-steady here
degradation records	k_0, E_d, n fitted, not assumed	Liu ch. 5	lifetime; MP10.2B
button-cell fit	i_0 and ASR from measured curves	Incropera ch. 1	validation
daily schedule	the hourly price as driver	Liu ch. 5	grid operation; MP10.2B, L12

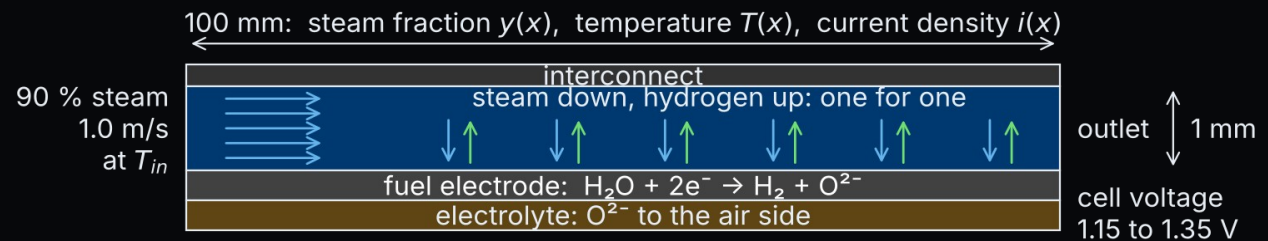
WHY THEY ARE LEFT OUT

Ex_10.2 solves the coupled channel in one dimension, each operating point at its steady state, scored against a boundary-value solver, so none of them is needed. The first and third rows are the two mini projects; the last meets the grid of L12.

Exercise 10.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the electrolyser, its data, the three coupled laws
- **4** — the reference; finite volumes coupled by Newton-Raphson
- **5** — Part A: one PINN for every operating point; 1000 points timed
- **6** — Part B: the optimal voltage under the 10 K limit, by gradient
- **7 – 9** — the report; mini projects MP10.2A and MP10.2B



THE QUESTION EACH PART ANSWERS

1 – 3	which laws couple, and how?
4	what is the coupling worth?
5	one network, every point?
6	which voltage, within 10 K?

REPORT QUESTION, TO CONCLUDE

For which task in this notebook would you use finite volumes, for which the network - and what changes when a voltage must be chosen for every hour of a day? Say which of your numbers rest on estimates.

SUMMARY

Key Takeaways

SOLID OXIDE CELL

what it is	a ceramic cell, two directions
function	steam to hydrogen, heat helping
train, tune	operated at or near V_{tn}
vs low temperature	a quarter less electricity
used for	storing surplus power as fuel

LIFETIME OPTIMISATION

what it is	a PINN over voltage and inlet heat
function	three fields, the cost, its gradient
train, tune	once: 1.15 – 1.35 V, 750 – 850 °C
vs a solver	autograd gradient; a solver re-solves
used for	many points, and the choice of V

Next — L11.1: a network in a real-time loop, where there is no PDE at all.

Questions

1 – 2 · A NETWORK FOR THE FUEL CHANNEL

1 **How can one network give the steam, the temperature and the current along the channel at every operating point?**

It takes the position, the cell voltage and the inlet temperature and returns the three fields, trained over the whole operating range until the three laws hold. The inlet values are built in. Any operating point is then read off, not solved again. *Go deeper: the network and its inputs (Ex_10.2 section 5); Liu ch. 2.2.*

2 **Why does the local current make the channel harder to solve, and how is the network checked?**

The current has no equation of its own: at every point it is the root of the charge transfer at the local steam fraction and temperature. The network takes it as a third residual, in units of 50 mV. Against a reference solver it was within 0.0022 and 0.19 K. *Go deeper: the coupled laws (Ex_10.2 sections 3 – 5); the comparison with finite volumes.*

3 – 4 · THE OPTIMAL VOLTAGE

3 **How is the optimal cell voltage found with the trained network?**

The cost of an hour, the electricity less the hydrogen plus the ageing, is read from the fields, with the 10 K limit as a penalty. Autograd gives its derivative in the voltage, and gradient descent finds 1.250 V in seconds; a sweep of 81 solver runs agrees. *Go deeper: with and without the limit (Ex_10.2 section 6); Liu ch. 5.*

4 **Why does a network help to plan the cell's operation over its lifetime, and what should not be claimed?**

Over a lifetime the decision is a path of voltage and temperature, and a solver's gradient costs one solve per variable. A network gives every gradient by autograd. The ageing constants are assumed, so a predicted lifetime ranks schedules; it is not a promise. *Go deeper: Liu ch. 5; the spread of published ageing rates.*

NEXT

L11.1 — Vision-Based Navigation

A network in a real-time control loop on a robot car, where the model is learned from camera frames and there is no PDE at all.

BEFORE THE NEXT SESSION

read	Liu ch. 5
run	Ex_10.2, sections 1 to 6
bring	your optimal voltage, and its worth
think about	what was physics, what was price

Block ten is finished: lithium in a particle, and steam in a ceramic cell - the course's last partial differential equations.